# Bill of Materials & Technical Report

**ELECTRIC FIXED-WING RC AIRCRAFT (1.1 M WINGSPAN)**

## 1. Project Brief & System Design Envelope

This report presents the complete engineering Bill of Materials (BOM), trade-off analysis, structural rationale, and propulsion sufficiency verification for a high-wing electric RC trainer aircraft designed to comply with all operational parameters.

AIRCRAFT BRIEF CONSTRAINT

**1.10 m Wingspan**

High-wing configuration with Clark-Y flat-bottom airfoil for high and stall tolerance. 2A digital protocol with dual-aileron control.

PROPULSION & CONTROL

**Single Electric Motor / 4-CH**

Brushless outrunner motor (980 KV), 10x4.7 prop, 2.4 GHz AFHDS aerodynamic stability

AIRFRAME & MATERIAL

**5mm Fluted PP + Carbon**

Lightweight, impact-resistant Coroplast reinforced by a 6mm 3K repairable with CA/hot glue.

TAKEOFF & LANDING CAPABILITY

**Ground Roll & Hand Launch**

Dual takeoff capability via duralumin main gear with 55mm EVA carbon spar; highly wheels or direct hand launch at full throttle.

## 2. Part A & B — Component Selection & BOM (Powerplant & Flight Power)

| Component | Recommendation 1 (Primary / Performance) | Recommendation 2 (Alternative / Budget) | Key Specifications | Qty | Cost (₹) |
|---|---|---|---|---|---|
| **Electric Motor** | **SunnySky X2212 V3 980KV**<br>Source: Robu.in / QuadStore | A2212 / 2212 1000KV Outrunner<br>Source: Flyware India | 980 KV, 18.5A Max Current, 205W, 3.17mm Shaft, 52g | 1 | **₹1,350**<br>(Alt: ₹480) |
| **Speed Controller (ESC)** | **Hobbywing FlyFun 30A V2**<br>Source: Robu.in / RCMumbai | Yellow / Flycolor 30A ESC<br>Source: Amazon India | 30A Cont. (40A Burst), 5V/3A Switch BEC, 2S-4S, 32g | 1 | **₹1,350**<br>(Alt: ₹480) |
| **Flight Battery** | **Gens Ace / Bonka 3S 2200mAh 45C**<br>Source: Robu.in | Orange 3S 2200mAh 30C LiPo<br>Source: QuadStore | 3S1P (11.1V Nominal), 2200mAh, 45C Discharge, XT60, 185g | 1 | **₹1,750**<br>(Alt: ₹1,350) |
| **Propeller** | **APC 10x4.7 Slow Fly (SF)**<br>Source: Robu.in | Gemfan 1045 Nylon Propeller<br>Source: Amazon India | 10" Dia, 4.7" Pitch, 2Blade Composite, 3.17mm Bore, 15g | 1 pair | **₹380**<br>(Alt: ₹150) |

**System Selection Strategy Note**

Recommendation 1 components prioritize high mechanical tolerance, efficient switching power conversion (SBEC), and thermal reliability during continuous operation. Recommendation 2 provides an entry-level functional baseline at reduced cost.

## 2. Part A & B — Component BOM (Radio, Control, Airframe & Hardware)

| Component | Recommendation 1 (Primary / Performance) | Recommendation 2 (Alternative / Budget) | Key Specifications | Qty | Cost (₹) |
|---|---|---|---|---|---|
| **Radio System (TX & RX)** | **FlySky FS-i6X 2.4GHz 10CH + FSiA6B** Source: Robu.in | FlySky FS-i6 6CH + FS-iA6 Source: Amazon India | 2.4GHz AFHDS 2A, PWM/PPM/i-BUS, Range >1000m | 1 set | ₹4,600 (Alt: ₹3,950) |
| **Control Servos** | **Emax ES08MA II Metal Gear 12g** Source: Robu.in / QuadStore | TowerPro SG90 9g Nylon Gear Source: Local RC Shop | Torque: 2.0 kg·cm @ 4.8V, Speed: 0.12s/60°, Dual Ball Bearing | 4 | ₹1,160 (Alt: ₹320) |
| **Airframe & Structural Spar** | **5mm Fluted PP + 6mm Carbon Tube** Source: Local Plastics + Robu.in | 5mm Depron Foam Board + Wood Source: Stationers | Coroplast Density 400g/m², 3K Roll Wrapped Carbon Spar 600mm | 1 set | ₹650 (Alt: ₹350) |
| **Landing Gear & Wheels** | **Duralumin Aluminum + 55mm EVA** Source: Robu.in | 2.5mm Steel Wire + Sponge Wheels Source: DIY Fabrication | Main landing gear height 110mm, 55mm EVA foam wheels, steerable tail skid | 1 set | ₹450 (Alt: ₹250) |
| **Wiring & Connectors** | **Amass XT60H + 3.5mm Bullets + Wire** Source: Robu.in | Generic XT60 + Standard Wires Source: Local Electronics | 14 AWG Silicone Wire, Gold-plated 3.5mm Bullet connectors, 2x Servo Extensions | 1 set | ₹320 (Alt: ₹220) |
| **Adhesives & Hardware** | **5-Min Epoxy + Hot Glue + Ball Links** Source: Hardware Store / Robu | Standard Glue Sticks + Horns Source: Craft Store | 1.2mm Pushrods in Teflon sleeves, Nylon control horns, Hightemp glue sticks | 1 set | ₹550 (Alt: ₹350) |

## 3. Part C — Engineering Rationale & Selection Justification

**C.1 Motor–Propeller Matching & Thrust Stand Verification:** For a high-wing trainer operating at moderate cruise velocities (10–14 m/s), low-KV motors paired with high-diameter, shallow-pitch propellers maximize static thrust and acceleration efficiency rather than top speed.

**Motor Power Input Equation:**

$$P_{in} = V_{nominal} \times I_{max} = 11.1\ V \times 18.5\ A = 205.35\ W$$

At 980 KV on 3S LiPo (~10,870 no-load RPM), an APC 10x4.7 SF propeller yields a full-throttle motor thrust of **880 g (8.63 N)** at 18.5 A draw.




$$\text{Current Margin} = [(I_{ESC} - I_{Max}) / I_{Max}] \times 100\% = [(30\ A - 18.5\ A) / 18.5\ A] \times 100\% = 62.16\%$$

**C.2 ESC Rating & Electrical Safety Margin Calculation:** The maximum continuous current draw of the SunnySky X2212 motor under 100% static throttle with a 10x4.7 prop is 18.5 A. Choosing a 30 A continuous rated ESC provides an operational current headroom margin calculated as:

## 3. Part C — Engineering Selection Justifications (Continued)

**C.3 Battery Selection & Operational Trade-Offs:** A 3S (11.1V nominal) 2200 mAh LiPo battery was selected. The continuous discharge capacity at 45C rating is:

$$I_{max,discharge} = Capacity \times C\text{-}Rating = 2.2\ Ah \times 45\ C = 99.0\ A$$

Since peak motor draw is 18.5 A, the battery operates at under 19% of its max discharge limit, preventing thermal degradation and voltage drop under load. A 1500 mAh pack saves 45 g but limits endurance to ~8 mins; 2200 mAh provides ~14 mins cruise time while correctly balancing the nose Center of Gravity (C.G. at 28% MAC).

**C.4 Control Surface Allocation & Servo Aerodynamic Torque:** Four independent micro-servos are deployed: Aileron Left, Aileron Right, Elevator, and Rudder/Tailwheel. The dual-aileron setup enables flaperon flight mixing.

**Hinge Torque Calculation at $V_{max}$ (18 m/s):**

$$T_{hinge} = 0.5 \times \rho \times V^2 \times C_{surface}^{\ 2} \times L_{surface} \times \sin(\delta) = 0.42\ kgcm$$

Emax ES08MA II servos (2.0 kg·cm torque) deliver a **4.76× safety factor** over peak aerodynamic blowback torque.

**C.5 Structural Material Selection Matrix:** Evaluation of primary airframe materials based on structural density, stiffness, cost, and field repairability:

| Material Property | 5mm Fluted PP (Coroplast) | 5mm Depron Foam Board | Balsa Wood & Plywood |
|---|---|---|---|
| **Flexural Rigidity** | High along flutes; needs carbon spar | Moderate; brittle under bend | Very high grain directional strength |
| **Impact Durability** | **Excellent** (Absorbs high kinetic energy) | Poor (Cracks/fractures on impact) | Moderate (Splinters upon nose crash) |
| **Mass Density** | ~400 g/m² (Airframe ~220 g) | ~200 g/m² (Airframe ~160 g) | Variable (Airframe ~280 g) |
| **Field Repairability** | **Instant** (Hot glue / CA + Activator) | Requires Foam-safe Glue (Slow cure) | Requires Epoxy / CA gluing + skinning |
| **Material Cost** | **Lowest** (~₹150 / sheet) | Moderate (~₹350 / sheet) | High (~₹1,200 total wood stock) |

**C.6 Electrical & Mechanical Integration Compatibility:**

- **RF Protocol:** FlySky FS-i6X uses AFHDS 2A frequency-hopping 2.4GHz protocol, native to the included FS-iA6B receiver with dual-antenna diversity.
- **Power & Mechanical Interfacing:** Battery XT60 male couples to ESC XT60 female. Motor 3.5mm bullet leads mate to ESC phase outputs. Motor 16x19mm bolt pattern matches the 3mm plywood firewall mount.

## 4. Part D — Budget Analysis & Competition Strategy

Generic 2212 motor, yellow 30A ESC, SG90 nylon servos, 30C battery, Depron DIY frame. Minimal financial risk.

**PERFORMANCE BUILD TOTAL**

**BUDGET-ORIENTED BUILD TOTAL**

**₹ 8,180 INR**

**₹ 13,010 INR**

SunnySky V3 motor, Hobbywing 3A SBEC ESC, Emax metal gear servos, 45C LiPo, carbon reinforced PP.

**D.2 Safe Cost-Saving Strategies:** 1) *Hybrid Servo Deployment:* Metal gear servos on high-impact Elevator/Rudder surfaces and nylon servos on Ailerons saves ₹400. 2) *In-House Airframe Fabrication:* Hand-cutting 5mm fluted PP sheet using downloadable CAD templates saves ~₹1,800 vs pre-cut kits. 3) *Reusable Ground Equipment:* The radio system (₹4,600) is a capital investment reusable across future aircraft models.

**D.4 Competition Selection & Justification:** The **Performance-Oriented Configuration (₹13,010)** is selected for competition. The Hobbywing ESC's 3A Switch-mode BEC eliminates receiver brownout risks during multi-servo maneuvers, while metal-gear servos prevent gear strip failures during ground-roll landing impacts.

## 5. Takeoff Mass & Propulsion Sufficiency Analysis

| Component / Sub-System Unit | Mass (grams) | Mass Share (%) |
|---|---|---|
| Airframe Structure (Fluted PP sheet + Carbon Spar + Hardware) | 220 g | 34.5% |
| Motor Unit (SunnySky X2212 980KV + Prop Adapter + Mount) | 52 g | 8.2% |
| Speed Controller (Hobbywing 30A V2 with leads) | 32 g | 5.0% |
| Flight Battery (Gens Ace 3S 2200mAh 45C LiPo) | 185 g | 29.0% |
| Servos (4x Emax ES08MA II @ 12g each) | 48 g | 7.5% |
| Receiver (FlySky FS-iA6B with antennas) | 12 g | 1.9% |
| Propeller & Landing Gear Assembly (EVA wheels + Duralumin) | 60 g | 9.4% |
| Wiring, Pushrods, Horns & Fasteners | 28 g | 4.5% |
| **TOTAL ALL-UP WEIGHT (AUW)** | **637 g** | **100.0%** |

**Aerodynamic & Propulsion Metric Calculations:**

- **Wing Area (S):** 1.10 m × 0.18 m = 0.198 m² = 19.8 dm²
- **Wing Loading (WL):** m / S = 637 g / 19.8 dm² = **32.17 g/dm²** (Optimal trainer range: 25–40 g/dm²)
- **Stall Speed ($V_{stall}$):** √[ (2 × m × g) / (ρ × S × $C_{L,max}$) ] = √[ (2 × 0.637 × 9.81) / (1.18 × 0.198 × 1.2) ] = **6.68 m/s (24.0 km/h)**
- **Thrust-to-Weight Ratio (T/W):** 880 g Thrust / 637 g AUW = **1.38 : 1** (Exceeds 1.0 limit; hand launch capable)
- **Power-to-Weight Ratio (P/W):** 205.35 W / 0.637 kg = **322.4 W/kg (146.2 W/lb)** (Sport-aerobatic power density)
- **Pitch Speed ($V_{pitch}$):** (8158 RPM / 60) × 0.1194 m = 16.23 m/s (58.4 km/h) → **$V_{pitch}$ / $V_{stall}$ = 2.43** (>1.5 required)

**Final Engineering Sufficiency Verdict**

The propulsion configuration delivers **880 g static thrust** against an estimated **637 g AUW**. With a **T/W ratio of 1.38**, a **322.4 W/kg power density**, and a pitch speed 2.43× stall speed, propulsion is fully sufficient for hand-launch climbouts, wind resistance up to 20 km/h, and safe recovery operations.